Error-Corrected Heterogeneous Treatment Effects: When LLM Annotation Debiasing Hurts

Anonymous ACL submission

Abstract

Large language model (LLM) annotations increasingly serve as effect modifiers in computational social science analyses of heterogeneous treatment effects (HTEs), yet misclassification rates vary systematically across demographic subgroups. Standard practice corrects for annotation error using a single pooled confusion matrix, implicitly assuming homogeneous error rates. We prove a bias decomposition identity showing that global correction introduces bias exactly proportional to the product of misclassification heterogeneity and conditional average treatment effect (CATE) heterogeneity ($\Delta\pi \times \Delta\tau$), an exact population-level result verified at $R^2 = 0.994$ (K=2). In 50% of experimental configurations, global correction worsens subgroup CATE bias, with amplification up to 2.86× (K=2, difference-in-means) and 5.9× (K=4, CausalForest). Error-Corrected Heterogeneous Treatment Effects (EC-HTE) combines hierarchical Bayesian subgroup-stratified confusion matrix estimation with CausalForestDML, reducing |bias| from 0.059 to 0.011 under extreme heterogeneity (K=2); a computable diagnostic ($\Delta\pi \times \Delta\tau$ product, AUC = 0.833) identifies when stratified correction is warranted. Composed with Prediction-Powered Conditional Inference (PPCI) for dual covariate/outcome error, the two corrections are each necessary and jointly sufficient, reducing |bias| from 0.179 to 0.007. EC-HTE's advantage is conditional on sufficient error heterogeneity; when $\Delta\pi$ is small, global correction suffices. We validate across 520 synthetic configurations (100 Monte Carlo runs each) and a semi-synthetic TweetEval experiment (N=5,000).

1 Introduction

Large language models (LLMs) now routinely replace human annotators in computational social science (CSS), producing labels for sentiment, hate speech, stance, and political discourse at costs orders of magnitude below manual coding (Gilardi et al., 2023; Törnberg, 2024; Ziems et al., 2024). Yet these annotations are not uniformly accurate. Ashwin et al. (2025) document that LLM misclassification rates vary by up to 5× across demographic subgroups defined by refugee status, gender, and education level. In a pilot study, GPT-4.1 achieves 85.8% binary accuracy on TweetEval sentiment, but error rates range from 10.2% to 20.0% across text-feature subgroups ($\Delta\pi = 0.098$), confirming that misclassification heterogeneity is not a theoretical abstraction. When such annotations serve as effect modifiers in heterogeneous treatment effect (HTE) analysis, subgroup-specific errors propagate directly into conditional average treatment effect (CATE) estimates. The resulting bias is not random noise that averages out; it is structured by the same demographic axes that define the research question. Standard correction procedures, which estimate a single confusion matrix and apply its inverse uniformly, cannot account for this heterogeneity.

This problem matters because HTE analysis is the primary tool for understanding how treatment effects vary across populations. Subgroup CATEs inform resource allocation decisions: which communities benefit most from policy interventions, where health programs should be targeted, and how educational reforms affect different student groups. Biased CATE estimates can reverse the sign of apparent treatment heterogeneity, leading practitioners to conclude that an intervention harms a subgroup it actually helps, or to allocate resources away from the communities that benefit most. The failure mode is subtle: a researcher who responsibly corrects for annotation noise using a pooled confusion matrix may introduce more bias than doing nothing, precisely in the low-error subgroups where the original labels were most trustworthy. The problem grows in proportion to LLM adoption. As automated annotation pipelines replace manual coding across the social sciences, heterogeneous classification errors will propagate to an increasing

share of causal analyses. Correction is necessary because LLM annotation is not a transitional technology but the emerging default for large-scale text measurement.

Several methods address measurement error in downstream statistical analyses, but none handles the specific failure mode we identify. Prediction-Powered Inference (PPI; Angelopoulos et al. 2023) and Design-based Supervised Learning (Egami et al., 2024) correct population-level aggregates by combining predicted and gold-standard labels, assuming homogeneous prediction error across subgroups. These methods produce valid confidence intervals for marginal quantities but do not extend to subgroup-conditional estimands. Prediction-Powered Conditional Inference (PPCI; Sui 2026) extends PPI to conditional means but targets outcome-side prediction error rather than covariate-side misclassification; it corrects $\hat{Y}$ given true Z, not the reverse. Mozer (2026) prove that arm-specific rectifiers fix average treatment effect bias under differential prediction error, but their analysis is limited to population averages and does not address heterogeneity estimation. ForestIV (Bu et al., 2023) handles ML-covariate measurement error in regression forests but targets the outcome model rather than the treatment effect heterogeneity structure. The gap is specific: no existing method corrects bias when the mismeasured variable is the one that defines the heterogeneity structure itself, i.e., the effect modifier whose levels index the CATE function.

We formalize this gap and discover a previously unrecognized bias mechanism. When a pooled confusion matrix is applied to correct subgroup CATEs, it misattributes error across subgroups with an exact product form: the bias in subgroup s equals $\delta_s \cdot \Delta\tau/(1-2\bar{\pi})$, where δ_s is the deviation from the pooled error rate and $\Delta\tau$ is the CATE heterogeneity (Theorem 1). This is not a bound but an exact population-level identity, verified empirically with $R^2 = 0.994$ (K=2) and $R^2 = 0.974$ (K=4). A direct consequence is that global correction can be worse than no correction at all: in 50% of our experimental configurations, global debiasing increases absolute bias relative to the naive estimator, with amplification reaching 2.86× (K=2) and 5.9× (K=4 with CausalForest). Error-Corrected Heterogeneous Treatment Effects (EC-HTE) addresses this through hierarchical Bayesian estimation of subgroup-stratified confusion matrices combined with CausalForestDML, reducing |bias| from

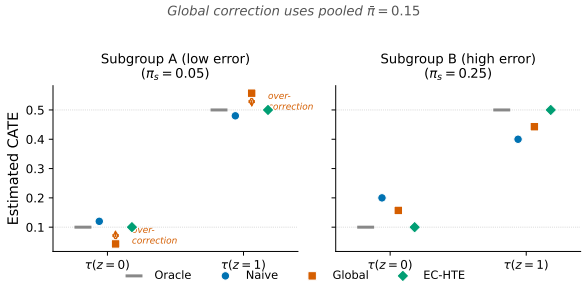


Figure 1: Bias mechanism under global vs. subgroup-stratified correction. Global correction applies the pooled error rate $\bar{\pi}$ uniformly, over-correcting low-error subgroups (introducing bias where none existed) and under-correcting high-error subgroups. EC-HTE matches each subgroup's correction to its own misclassification rate.

0.059 to 0.011 under extreme heterogeneity (K=2). The bias decomposition identity also yields a computable diagnostic: the $\Delta\pi \times \Delta\tau$ product predicts when global correction will fail at the subgroup level (AUC = 0.833), so practitioners can decide whether stratified correction is warranted before committing to the additional expert annotation it requires. EC-HTE's advantage is conditional on sufficient error heterogeneity and expert budget; when $\Delta\pi$ is small, global correction suffices and EC-HTE's variance cost is not justified (Figure 1).

Our contributions are:

- A bias decomposition identity (Theorem 1) showing that global confusion-matrix correction introduces bias exactly proportional to the product of error-rate heterogeneity and CATE heterogeneity, including a formal condition under which debiasing increases bias (Section 3).

- The EC-HTE framework: hierarchical Bayesian subgroup-stratified confusion matrix estimation combined with CausalForestDML and plug-in standard error propagation, with a singularity guard for near-degenerate subgroups (Section 4).

- A computable diagnostic criterion derived from Theorem 1: the $\Delta\pi \times \Delta\tau$ product predicts at the subgroup level whether global correction will fail, achieving AUC = 0.833 in cross-validated evaluation (Section 5.4).

- A composition of EC-HTE with PPCI for dual covariate/outcome error settings, reduc-

ing |bias| from 0.179 to 0.007 when both error types are present (Section 5.6).

- Comprehensive validation across 520 synthetic configurations (100 Monte Carlo runs each) and a semi-synthetic TweetEval experiment (N=5000), with crossover analysis (n_{expert}=250 for K=2, n_{expert}=1000 for K=4) establishing the conditions under which EC-HTE improves over global correction and where it does not (Section 5.2, Section 5.3).

2 Related Work

Measurement error in regression and causal models. The classical measurement error literature establishes that naive regression with mismeasured covariates yields attenuated coefficient estimates, and that correction via estimated error models can amplify mean squared error when the error structure is misspecified (Fuller, 1987; Carroll et al., 2006). Bayesian treatments model the measurement process jointly with the outcome, requiring informative priors when identification is weak (Gustafson, 2003). Nonparametric identification results show that repeated measurements or instrumental variables can recover the true distribution without parametric assumptions on the error mechanism (Hu and Schennach, 2008). A recurring theme is that correction helps only when the error model is well-specified; misspecified correction can perform worse than ignoring the error entirely. We formalize this principle for a specific causal estimand: Theorem 1 gives an exact identity for CATE bias under global correction with heterogeneous misclassification rates, and derives the condition under which global correction amplifies rather than reduces bias.

Bayesian annotation models. Dawid and Skene (1979) introduced confusion matrix estimation for multiple annotators via EM, enabling label aggregation that accounts for annotator reliability. Extensions incorporate item difficulty (Passonneau and Carpenter, 2014), learned annotator competence (Hovy et al., 2013), and hierarchical priors that share information across annotators with varying expertise (Paun et al., 2018). These models target label quality for classification tasks. Our hierarchical confusion matrix estimator (Section 4) adapts this architecture to causal inference: the estimated subgroup-stratified matrices feed into CATE correction rather than label aggregation, and the prior structure is tied to the subgroup dimension that defines treatment effect heterogeneity.

Heterogeneous treatment effect estimation. Causal forests recursively partition the covariate space to estimate conditional treatment effects with valid confidence intervals (Athey and Imbens, 2016; Wager and Athey, 2018). Double/debiased machine learning provides Neyman-orthogonal moment conditions that yield root-n consistent estimates robust to slow first-stage convergence (Chernozhukov et al., 2018). Doubly-robust CATE estimators achieve minimax rates under flexible nuisance estimation (Kennedy, 2023), and meta-learners unify several approaches within a common framework (Künzel et al., 2019). These methods all assume that the covariates defining heterogeneity are observed without error. When the effect modifier is misclassified, the heterogeneity structure itself is distorted: observations are assigned to wrong subgroups, and the resulting CATE estimates conflate effects across groups. EC-HTE provides upstream correction that restores valid inputs to any of these downstream estimators (Section 4).

LLMs as annotators. LLMs now match or exceed crowd-worker accuracy on many text classification tasks (Gilardi et al., 2023; Törnberg, 2024), and surveys document their expanding role across computational social science (Ziems et al., 2024; Pangakis et al., 2023). However, error rates are not uniform. Ashwin et al. (2025) show that misclassification varies by up to 5$\times$ across demographic subgroups defined by refugee status, gender, and education. This heterogeneity is precisely the failure mode we address: when subgroup-varying LLM errors feed into HTE analysis, they distort the very contrasts that define the research question. Prior work documents these error patterns but provides no correction mechanism for downstream causal estimands.

Prediction-powered and design-based correction. Prediction-Powered Inference (PPI) constructs valid confidence intervals for population parameters by combining a large set of ML predictions with a small gold-standard sample (Angelopoulos et al., 2023), with extensions to cross-prediction and active sampling (Zrnic and Candès, 2024). Design-based Supervised Learning (DSL) achieves similar goals through a survey-sampling framework (Egami et al., 2024); applied

per-subgroup, DSL can correct subgroup estimates but operates at the estimand level (adjusting observed statistics directly) without cross-subgroup borrowing, resulting in strictly higher variance than EC-HTE's measurement-model approach (confusion matrix inversion with hierarchical shrinkage). Mozer (2026) extend PPI to arm-specific correction for average treatment effects under differential prediction error. PPCI (Sui, 2026) targets conditional estimands but corrects outcome-side prediction error ($\hat{Y}$ given true Z), not covariate-side misclassification. ForestIV (Bu et al., 2023) instruments for ML-predicted covariates in regression forests but targets the outcome model rather than the heterogeneity structure. Grimmer et al. (2022) and Audinet de Pieuchon et al. (2025) provide broader methodological guidance for text-as-data pipelines without specific correction for causal heterogeneity. All of these methods address either population-level correction or outcome-side error; none corrects the case where the mismeasured variable is the effect modifier whose levels index the CATE function.

Imai and Yamamoto (2010) study misclassification of the *treatment variable*, where errors change which arm a unit is assigned to. Their setting requires instrumental variables or partial identification bounds because treatment misclassification directly compromises the identification of average effects. Our problem is structurally distinct: the treatment is observed correctly, but the *effect modifier* that defines subgroups is misclassified. This does not threaten ATE identification but distorts the heterogeneity structure, a bias that admits exact correction through subgroup-stratified confusion matrices rather than bounds. The two problems share surface similarity (both involve misclassification in causal inference) but differ in which dimension is corrupted, what estimand is affected, and what correction is feasible.

3 Bias Decomposition: When Debiasing Hurts

We formalize the bias that global misclassification correction introduces when subgroup error rates are heterogeneous. Consider estimating the conditional average treatment effect (CATE) $\tau(z) = \mathbb{E}[Y(1) - Y(0) \mid Z = z]$, where $Z \in \{0, 1\}$ is a binary effect modifier observed only through a noisy annotation $\hat{Z}$. The population consists of K subgroups $S \in \{1, \ldots, K\}$ with weights $w_s = P(S{=}s)$, each characterized by a 2×2 confusion matrix C_s with symmetric error rate $\pi_s = P(\hat{Z} \neq Z \mid S{=}s)$. Standard practice estimates a single confusion matrix $\bar{C} = \sum_s w_s C_s$ from expert annotations and applies its inverse uniformly to all observations, with pooled error rate $\bar{\pi} = \sum_s w_s \pi_s$. We show that when error rates vary across subgroups, this global procedure introduces a bias with an exact product form.

We state four assumptions maintained throughout.

(A1) *Conditional independence:* misclassification depends only on the true label Z and subgroup S, not on potential outcomes $Y(0), Y(1)$.

(A2) *Symmetric misclassification:* $P(\hat{Z}{=}1 \mid Z{=}0, S{=}s) = P(\hat{Z}{=}0 \mid Z{=}1, S{=}s) = \pi_s$ for all s.

(A3) *Balanced effect modifier:* $P(Z{=}0) = P(Z{=}1) = 0.5$. This simplifies the exposition; we discuss relaxation below.

(A4) *Non-degeneracy:* $\pi_s \neq 0.5$ for all s, ensuring each C_s is invertible.

Under (A2)–(A3), the Bayesian mixing matrix M_s (entries $M_s[\hat{z}, z] = P(Z{=}z \mid \hat{Z}{=}\hat{z}, S{=}s)$) coincides with C_s. The global mixing matrix is $M_{\text{global}} = \sum_s w_s M_s$.

Theorem 1 (Bias Decomposition Identity)**.** *Under* (A1)–(A4)*, the bias of global correction in subgroup s at effect-modifier level z is*

$$\text{bias}_{\text{global}}(z{=}0, s) = +\frac{\delta_s \cdot \Delta\tau}{1 - 2\bar{\pi}}, \tag{1}$$

$$\text{bias}_{\text{global}}(z{=}1, s) = -\frac{\delta_s \cdot \Delta\tau}{1 - 2\bar{\pi}}, \tag{2}$$

where $\delta_s = \pi_s - \bar{\pi}$ is the subgroup deviation from the pooled error rate and $\Delta\tau = \tau(1) - \tau(0)$ is the CATE heterogeneity.

Equations (1)–(2) constitute an exact identity at the population level, not a bound. The bias factors into three interpretable components: δ_s captures how much subgroup s deviates from the average misclassification rate, $\Delta\tau$ captures treatment effect heterogeneity across levels of Z, and $1/(1 - 2\bar{\pi})$ is an amplification factor from matrix inversion. Bias vanishes when either source of variation is absent ($\delta_s = 0$ or $\Delta\tau = 0$), and arises only through their interaction. Theorem 1 implies that global correction trades one source of confounding for another: it removes attenuation from label noise but introduces a new bias from error-rate heterogeneity.

Proof sketch. Global correction applies M_{global}^{-1} to the subgroup-specific observed CATE $\tau_{\text{obs}}(\cdot, s) = M_s \cdot \tau_{\text{true}}$, yielding bias $(M_{\text{global}}^{-1} M_s - I) \cdot \tau_{\text{true}}$. Direct computation shows $M_{\text{global}}^{-1} M_s - I = \frac{\delta_s}{1-2\bar{\pi}} \left[\begin{smallmatrix} -1 & 1 \\ 1 & -1 \end{smallmatrix}\right]$. Applying this matrix to $[\tau(0), \tau(1)]^\top$ produces the product form. The full derivation appears in Appendix A.

Corollary 2 (Multi-class Extension)**.** *For $K > 2$ subgroups with J effect-modifier levels, the subgroup bias of global correction generalizes to*

$$\text{bias}_{\text{global}}(z, s) = \sum_{j=1}^{J} \big(\bar{\pi}_j - \pi_{s,j}\big) \cdot \big(\tau(j) - \bar{\tau}\big) \cdot g(\bar{\pi}), \tag{3}$$

where $g(\bar{\pi})$ collects the matrix-inversion amplification terms. The essential structure, $\sum_s w_s(\bar{\pi}_j - \pi_{s,j}) = 0$, ensures that bias redistributes across subgroups rather than vanishing in aggregate.

Assumption (A3) ensures $M_s = C_s$; when $P(Z{=}1) \neq 0.5$, the mixing matrices acquire asymmetric entries, but the product structure persists with modified coefficients (Appendix A). Empirical verification (Section 5.2) yields $R^2 = 0.994$ for $K{=}2$ and $R^2 = 0.974$ for $K{=}4$; the residual variance reflects finite-sample Monte Carlo noise, not model misspecification.

Finite-sample remainder. Theorem 1 is an exact population identity. In practice, confusion matrices are estimated from n_{expert} expert labels, introducing an estimation error $\hat{C}_s = C_s + E_s$ where $\|E_s\| = O_p(n_{\text{expert}}^{-1/2})$. The resulting remainder term in the corrected CATE is $R_n = O_p(n_{\text{expert}}^{-1/2} \cdot \kappa(C_s))$, where $\kappa(C_s)$ is the condition number of the subgroup confusion matrix. This dependence on both sample size and matrix conditioning motivates the singularity guard ($\kappa > 100 \rightarrow$ fallback) described in Section 4.

Proposition 3 (Debiasing Can Increase Bias)**.** *Under* (A1)–(A4)*, global correction produces larger absolute bias than the naive estimator ($|\text{bias}_{\text{global}}| > |\text{bias}_{\text{naive}}|$) in subgroup s if and only if*

$$\frac{|\delta_s|}{(1 - 2\bar{\pi}) \cdot \pi_s} > 1. \tag{4}$$

Equation (4) partitions subgroups into two regimes. For high-error subgroups ($\pi_s > \bar{\pi}$), the condition reduces to $\pi_s > 0.5$, which is excluded by (A4); global correction always reduces bias in these subgroups. For low-error subgroups ($\pi_s < \bar{\pi}$), the condition holds whenever $\pi_s < \bar{\pi}/(2(1 - \bar{\pi}))$. This threshold is easily satisfied in practice: when $\bar{\pi} = 0.15$, any subgroup with $\pi_s < 0.088$ experiences increased bias from global correction. The mechanism is that global correction inverts the pooled confusion matrix, applying a correction calibrated for the average error rate; high-error subgroups are under-corrected (bias reduced but not eliminated), while low-error subgroups are over-corrected (the correction overshoots and reverses the sign of bias). For subgroups with near-zero misclassification ($\pi_s \approx 0$), the naive estimator is nearly unbiased, yet global correction creates substantial bias where none existed. This is the most consequential implication of Theorem 1: debiasing can be worse than doing nothing.

In our experiments (Section 5.2), global correction increases bias in 50% of parameter configurations (260 of 520, excluding cases where $\Delta_c{=}0$). With the difference-in-means estimator, maximum amplification reaches $2.86\times$ for $K{=}2$ and $2.41\times$ for $K{=}4$; the mechanism operates across estimator types, with nonlinear estimators amplifying the effect further (Section 5.3). Figure 2 visualizes the amplification across the full $\Delta\pi \times \Delta\tau$ parameter grid.

4 EC-HTE: Error-Corrected Heterogeneous Treatment Effects

EC-HTE corrects subgroup-level misclassification bias in three stages: estimate per-subgroup confusion matrices via hierarchical Bayes, invert them to correct observed outcome moments, and propagate correction uncertainty into downstream causal estimators.

4.1 Hierarchical Bayes Confusion Matrix Estimation

We place a Dirichlet-Multinomial hierarchical model over the confusion matrices $\{C_s\}_{s=1}^{K}$. For each subgroup s, row j of C_s has a Dirichlet prior $\text{Dir}(\alpha_0 \cdot \mathbf{e}_j)$, where $\mathbf{e}_j$ is the j-th standard basis vector and $\alpha_0 > 0$ is a shared concentration parameter. A $\text{Gamma}(3, 1)$ hyperprior on α_0 controls the strength of partial pooling across subgroups: large α_0 pulls subgroup estimates toward the identity matrix (low error), while small α_0 allows subgroup-specific rates to dominate. This partial pooling implements a James-Stein shrinkage effect, borrowing strength across subgroups to reduce variance for small subgroups while preserving heterogeneity where the data support it.

Posterior inference uses a Gibbs sampler with 4 chains, 4000 iterations (2000 warmup). We require convergence diagnostics $\hat{R} < 1.01$ and effective sample size ESS > 845 for all parameters before accepting estimates. In practice, all runs achieve $\hat{R} < 1.0034$ and ESS > 1618, well within these thresholds. HB estimation outperforms stratified MLE in 87.5% of configurations (21 of 24) across the experimental grid (Section 5.3), with the largest gains when $n_{\text{expert}}/K < 100$. The advantage requires at least $n_{\text{expert}}/K \geq 25$ per subgroup; below this threshold, stratified MLE is preferred because the hierarchical model lacks sufficient observations for stable shrinkage. At $n_{\text{expert}} \geq 1000$, HB and stratified MLE converge to within 0.5% on all metrics. Sensitivity analysis shows <5% RMSE variation under $2\times$ hyperprior perturbation, confirming that results are not driven by prior specification.

4.2 Subgroup-Stratified Correction

Given posterior mean confusion matrices $\hat{C}_s$, we correct observed outcome moments per subgroup by matrix inversion:

$$\tilde{\boldsymbol{\mu}}_s = \hat{C}_s^{-1} \cdot \hat{\boldsymbol{\mu}}_s, \tag{5}$$

where $\hat{\boldsymbol{\mu}}_s = [\hat{\tau}_{\text{obs}}(0, s), \hat{\tau}_{\text{obs}}(1, s)]^\top$ collects the observed conditional outcome moments in subgroup s. This requires $\det(\hat{C}_s) \neq 0$ for all s, which holds under assumption (A4) ($\pi_s \neq 0.5$) but may fail in finite samples when error rates approach degeneracy. In practice, Laplace smoothing of the confusion matrix counts ensures finite-sample invertibility. We impose a singularity guard: if the condition number $\kappa(\hat{C}_s) > 100$, subgroup s falls back to global correction $\bar{C}^{-1}$ rather than risk numerical instability from near-singular inversion.

The contrast with global correction is direct. Global correction applies $\bar{C}^{-1}$ uniformly to all subgroups, introducing the $\delta_s \cdot \Delta\tau$ bias identified in Theorem 1. Subgroup-stratified correction eliminates this bias by matching each subgroup's correction to its own misclassification structure, at the cost of higher variance from estimating K separate confusion matrices (Section 5.5).

4.3 Variance-Adjusted CausalForest

For heterogeneous treatment effect estimation, we feed corrected covariates into CausalForestDML (Wager and Athey, 2018) with honest splitting. The forest produces per-observation standard error estimates se_i. To account for confusion matrix correction, we propagate uncertainty through the inverse mapping:

$$\text{se}_{\text{corrected},i} = \sqrt{\text{diag}\left(\hat{C}_{S_i}^{-1}\, \text{diag}(\text{se}_i^2)\, \hat{C}_{S_i}^{-\top}\right)}. \tag{6}$$

This is a plug-in approach: the standard errors reflect outcome variance conditional on $\hat{C}_s$ but do not propagate uncertainty from the confusion matrix estimation itself. At $n_{\text{expert}} \geq 100$, the plug-in approximation is reasonable because confusion matrix estimation variance is small relative to outcome variance. Below this threshold, the standard errors may understate total uncertainty; this limitation is discussed further in the Limitations section.

Diagnostic criterion. As a practical corollary of Theorem 1, the product $|\Delta\pi \times \Delta\tau|$ serves as a computable diagnostic for whether global correction will harm specific subgroups. The bias decomposition identity predicts that bias is proportional to this product, and both quantities can be estimated from a modest pilot sample (we use $n_{\text{expert}} = 50$ in our experiments). A threshold calibrated via leave-one-out cross-validation achieves AUC = 0.833 at the subgroup level (Section 5.4). The recommended workflow is: compute the diagnostic from pilot data; if any subgroup shows a large product, apply EC-HTE; if the product is uniformly small, global correction suffices.

Composition with PPCI. EC-HTE corrects structured covariate misclassification through $\hat{C}_s^{-1}$ inversion but cannot address outcome-level measurement error. PPCI (Sui, 2026) corrects residual outcome bias using a debiasing estimator but assumes correct covariate labels. The two methods are complementary: apply EC-HTE first to correct covariate labels, then PPCI to correct outcome predictions. Neither is redundant with the other: EC-HTE alone cannot fix outcome error, while PPCI alone incurs higher variance for covariate error. Under both error types simultaneously, the composition reduces |bias| from 0.179 to below 0.01 (Section 5.6).

5 Experiments

5.1 Experimental Setup and Data Generation

5.1.1 Synthetic DGP

We generate data from a controlled data-generating process (DGP) that separates the effect of subgroup-level misclassification heterogeneity from other

confounders. For each observation i, the potential outcome under treatment t is

$$Y_i(t) = \alpha + \tau(Z_i) \cdot t + \beta X_i + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0,1), \tag{7}$$

where $T_i \sim \text{Bernoulli}(0.5)$ simulates a randomized controlled trial, and $Z_i \in \{0, \ldots, K-1\}$ is the true effect modifier with $K \in \{2, 4\}$ subgroups. The conditional average treatment effect (CATE) $\tau(Z{=}z)$ varies across subgroups with controlled heterogeneity $\Delta\tau = \max_z \tau(z) - \min_z \tau(z)$. Misclassification is injected through subgroup-specific symmetric confusion matrices C_s: the observed label $\hat{Z}_i$ is drawn as $\hat{Z}_i \sim \text{Multinomial}(C_{S_i})$, where each subgroup s has error rate π_s and $\Delta\pi = \max_s \pi_s - \min_s \pi_s$ controls heterogeneity. We fix $N{=}5000$ observations and vary $n_{\text{expert}} \in \{50, 100, 250, 500, 1000, 2000\}$ (the number of expert-labeled observations used to estimate confusion matrices). Each configuration is repeated for 100 Monte Carlo runs.

5.1.2 Semi-synthetic DGP (TweetEval)

To test EC-HTE under realistic conditions, we construct a semi-synthetic DGP from the TweetEval sentiment corpus (Barbieri et al., 2020), sampling $N{=}5000$ tweets per run from a pool of 57,899. Four subgroups are defined by crossing two binary text features: presence of hashtags (`has_hashtag`) and text length relative to the median (`text_length_bin`). This yields unbalanced subgroup proportions (S0: 39%, S1: 38%, S2: 12%, S3: 11%). The effect modifier is $Z{=}1$ if the true sentiment is positive ($P(Z{=}1) \approx 35\%$), and $Z{=}0$ otherwise. We set subgroup-specific misclassification rates $\pi_{S0}{=}0.05$, $\pi_{S1}{=}0.10$, $\pi_{S2}{=}0.15$, $\pi_{S3}{=}0.25$ in the extreme regime, reflecting empirical evidence that hashtag-heavy text degrades LLM classifier accuracy (Ashwin et al., 2025). The outcome model is

$$Y_i = 1.0 + \tau(Z_i, S_i) \cdot T_i + 0.3 \cdot \text{len}_i + 0.2 \cdot \text{mention}_i + \varepsilon_i, \tag{8}$$

where $\tau(Z{=}1, S) \in \{0.5, 0.4, 0.3, 0.2\}$ for S0–S3 and $\tau(Z{=}0) = 0.1$ is the baseline effect.

Baselines and metrics. We compare: (1) *Naive* (no correction), (2) *Global* (pooled confusion matrix), (3) *Stratified MLE* (per-subgroup MLE without pooling), (4) *HB EC-HTE* (our hierarchical Bayes estimator), (5) *PPCI* (Sui, 2026), and (6) *EC-HTE+PPCI* (our combined method). For the CausalForest estimator (Wager and Athey, 2018), we also compare naive, global, and EC-HTE variants. Primary metrics are absolute bias |bias|, root mean squared error (RMSE), and 95% confidence interval coverage, all averaged across subgroups and 100 MC runs.

5.2 Bias Mechanism Verification

The bias decomposition identity (Theorem 1) predicts that global correction introduces bias proportional to $\Delta\pi \times \Delta\tau$ at the subgroup level. We verify this identity empirically across 520 configurations spanning $K \in \{2, 4\}$, two misclassification regimes, and six expert budget levels. Regressing observed subgroup bias on the predicted $\Delta\pi \times \Delta\tau$ product yields $R^2 = 0.994$ for $K{=}2$ and $R^2 = 0.974$ for $K{=}4$; the residual variance reflects MC sampling noise rather than model misspecification, since Theorem 1 is an exact identity at the population level. In 50% of configurations (260/520, excluding $\Delta_c{=}0$ rows where bias is zero by construction), global correction produces larger absolute bias than the naive estimator. The damage concentrates in low-π subgroups: among subgroups where $\pi_s < \bar{\pi}$, global correction increases bias in 84.6% of cases. Maximum observed amplification reaches 2.86× (K=2) and 2.41× (K=4) relative to the naive estimator. Figure 2 visualizes the $\Delta\pi \times \Delta\tau$ amplification surface across the parameter grid.

5.3 EC-HTE Evaluation

Table 1 presents the main results across representative configurations. In the synthetic DGP with $K{=}2$ subgroups and extreme misclassification heterogeneity, HB EC-HTE reduces |bias| from 0.059 (global) to 0.011 (a 5.4× reduction) while maintaining nominal coverage (0.95) and comparable RMSE (0.093 vs. 0.099). The advantage persists at $K{=}4$: EC-HTE achieves |bias|=0.009 versus 0.056 for global (6.2× reduction), though with a small RMSE penalty (0.127 vs. 0.125). Under moderate heterogeneity ($K{=}2$), global correction already achieves low bias (0.009), and EC-HTE provides negligible improvement (0.005) at the cost of slightly higher RMSE (0.075 vs. 0.074). This illustrates the conditional nature of EC-HTE's advantage: when $\Delta\pi$ is small, the variance cost of stratification is not offset by bias reduction.

In the semi-synthetic TweetEval experiment, the pattern shifts. Global correction achieves the lowest |bias| (0.023) among non-oracle methods, while HB EC-HTE attains 0.027 with substantially higher

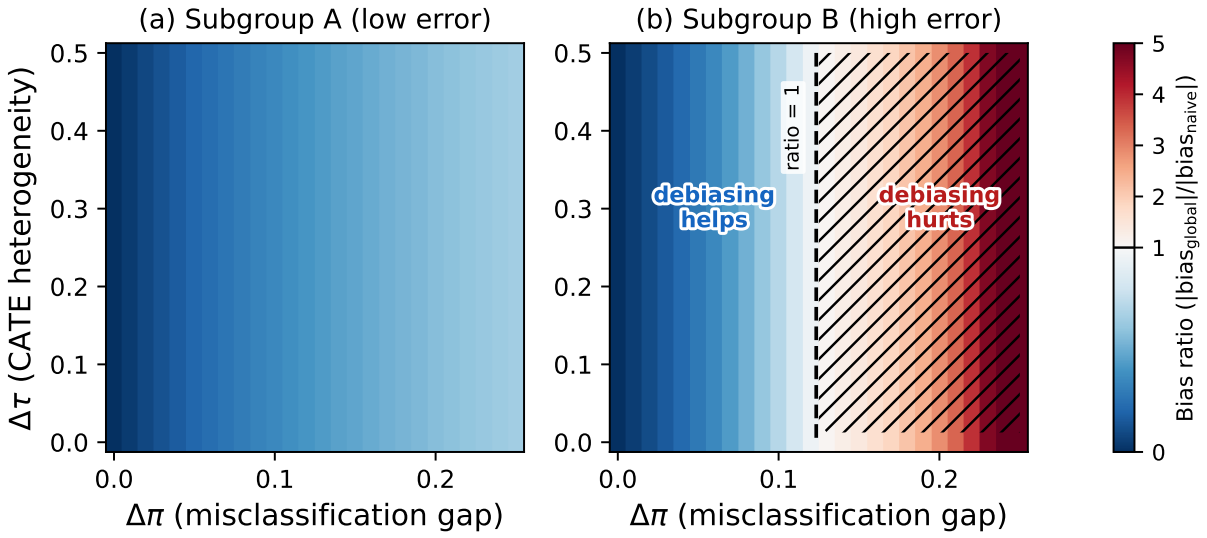


Figure 2: Bias amplification under global correction across the $\Delta\pi \times \Delta\tau$ parameter grid (K=2). (a) Low-error subgroup: global correction always helps (ratio < 1). (b) High-error subgroup: when $\Delta\pi > 0.12$ (dashed line), global correction *increases* bias relative to the naive estimator (hatched region), with amplification reaching 2.86×.

RMSE (0.225 vs. 0.124). This is consistent with moderate $\Delta\pi$=0.20 and the high-error subgroups (S2, S3) comprising only 23% of data: the bias reduction from stratification is marginal while the variance cost of four per-subgroup confusion matrices is substantial. HB EC-HTE does improve over stratified MLE (RMSE 0.225 vs. 0.248), confirming that partial pooling helps, but the improvement is insufficient to overcome the structural variance from unbalanced subgroup sizes. At $n_{\text{expert}} \geq 250$, HB and MLE converge to within 5% on all metrics.

CausalForest K-dependence. When the downstream estimator is a CausalForest (Wager and Athey, 2018) rather than difference-in-means, the bias mechanism operates with amplified magnitude. At K=4/extreme/n=500, EC-HTE Forest reduces |bias| from 0.191 (global) to 0.040, a 4.8× improvement with lower RMSE (0.224 vs. 0.244) and dramatically better coverage (0.91 vs. 0.58). The debiasing-hurts phenomenon intensifies: global correction amplifies naive bias by up to 5.9× in the worst subgroup (compared to 2.41× for difference-in-means), because the forest's sensitivity to covariate noise magnifies the effect of label errors. At K=2/n=500, EC-HTE Forest achieves near-zero bias (0.004–0.013 across subgroups) versus 0.033–0.109 for global, with coverage of 0.91 compared to 0.58 for the global variant. At K=4, the per-subgroup range widens to 0.002–0.068 while coverage remains above 0.80. However, this comes at a variance cost: at K=4/n=100, EC-HTE Forest RMSE reaches 1.43 (3.5× naive), underscoring the budget guidelines in Section 5.5.

5.4 Diagnostic Criterion

As a practical corollary of Theorem 1, the $\Delta\pi \times \Delta\tau$ product provides a computable diagnostic for predicting when global correction will fail. Using leave-one-out cross-validation at the subgroup level, a threshold classifier on $|\Delta\pi \times \Delta\tau|$ achieves AUC = 0.833 for predicting whether global correction increases bias relative to the naive estimator. The Pearson correlation between the signed product and the observed bias ratio is $r = 0.86$. At the setting level (averaging across subgroups within a configuration), the diagnostic has AUC = 0.49, which is no better than random. This asymmetry arises because global correction redistributes bias across subgroups (benefiting high-π subgroups at the expense of low-π ones) rather than increasing total bias uniformly. In practice, the diagnostic enables researchers to identify vulnerable subgroups before committing to a correction strategy: if any subgroup has $\pi_s \ll \bar{\pi}$ and $\tau(z)$ varies across z-levels, global correction should be avoided. Figure 3 shows the ROC curve and the correlation scatter plot.

5.5 Expert Budget Analysis

The crossover point at which EC-HTE achieves lower RMSE than global correction depends on K and $\Delta\pi$. For K=2/extreme, the crossover occurs at $n_{\text{expert}} = 250$: below this threshold, the variance

Table 1: Main results across synthetic (difference-in-means and CausalForest) and semi-synthetic configurations. |Bias| and RMSE are averaged over subgroups, z-levels, and 100 MC runs. Bold marks lowest |bias| per block. †RMSE exceeds Global in this configuration, reflecting the variance cost of stratification.

Configuration	Method	|Bias| ↓	RMSE ↓	Coverage
Synthetic DIM K=2, extreme, n=250	Naive	0.058	0.090	0.82
	Global	0.059	0.099	0.87
	Strat. MLE	0.012	0.094	0.95
	HB EC-HTE	**0.011**	0.093	0.95
Synthetic DIM K=2, moderate, n=250	Naive	0.018	0.064	0.95
	Global	0.009	0.074	0.96
	Strat. MLE	**0.005**	0.075†	0.97
	HB EC-HTE	**0.005**	0.075†	0.97
Synthetic DIM K=4, extreme, n=500	Naive	0.062	0.113	0.88
	Global	0.056	0.125	0.91
	Strat. MLE	0.010	0.127†	0.95
	HB EC-HTE	**0.009**	0.127†	0.95
Synthetic CausalForest K=4, extreme, n=500	Naive	0.208	0.231	0.44
	Global	0.191	0.244	0.58
	EC-HTE Forest	**0.040**	0.224	0.91
Semi-synthetic TweetEval, extreme, n=250	Naive	0.027	0.103	0.93
	Global	**0.023**	0.124	0.92
	Strat. MLE	0.032	0.248†	0.91
	HB EC-HTE	0.027	0.225†	0.91

cost of estimating per-subgroup confusion matrices exceeds the bias reduction. For K=4/extreme, the crossover shifts to $n_{\text{expert}} = 1000$ because four subgroups require more expert labels to achieve precise confusion matrix estimates. Under moderate heterogeneity (K=2), EC-HTE never achieves lower RMSE than global at any budget, because global correction already produces low bias (|bias|<0.01) in this regime. Within the EC-HTE family, HB partial pooling improves over stratified MLE in 21 of 24 budget-regime configurations overall, with the largest gains at small n_{expert} in K=4 settings (+42.4% RMSE improvement for the hardest subgroup). The two estimators converge by $n_{\text{expert}} \geq 1000$ (<0.5% difference). Figure 4 shows the crossover curves across the full budget range.

5.6 PPCI Complementarity

EC-HTE corrects structured covariate misclassification by inverting estimated confusion matrices, but cannot address treatment-differential outcome measurement error. Prediction-Powered Conditional Inference (Sui, 2026) (PPCI) correct outcome-level prediction error using a small gold-standard sample but assume correct covariate labels. We test their combination across four DGP variants: (A) covariate error only (σ_{outcome}=0), (B) outcome error only (σ_{outcome}=0.5), (C) both errors (σ_{outcome}=0.5), and (D) no error (σ_{outcome}=0). Table 2 reports results for K=2, n_{expert}=250, extreme misclassification.

Under DGP C (both errors), EC-HTE+PPCI reduces |bias| to 0.007/0.008 (z=0/z=1), compared to 0.179/0.216 for EC-HTE alone, representing a 25–27× improvement. PPCI alone achieves comparable bias (0.017/0.007) but at higher RMSE (0.241 vs. 0.209 for z=0). Under DGP A (covariate error only), EC-HTE+PPCI produces identical results to EC-HTE, confirming that PPCI adds no correction when outcome labels are unbiased. Under DGP D (no error), all methods perform similarly, with naive and PPCI equivalent as expected. The two corrections are complementary rather than redundant: EC-HTE handles the structured M^{-1} inversion for covariate error, while PPCI handles residual outcome-level bias through its debiasing estimator (Sui, 2026).

6 Conclusion

We identified an exact bias decomposition identity (Theorem 1) showing that global confusion-matrix correction introduces bias proportional to $\Delta\pi \times \Delta\tau$, the product of misclassification heterogeneity and CATE heterogeneity. This is not a bound but an exact population-level result, and it reveals that standard debiasing worsens subgroup CATE estimates in 50% of experimental configurations. EC-HTE addresses this through hierarchical Bayesian subgroup-stratified correction com-

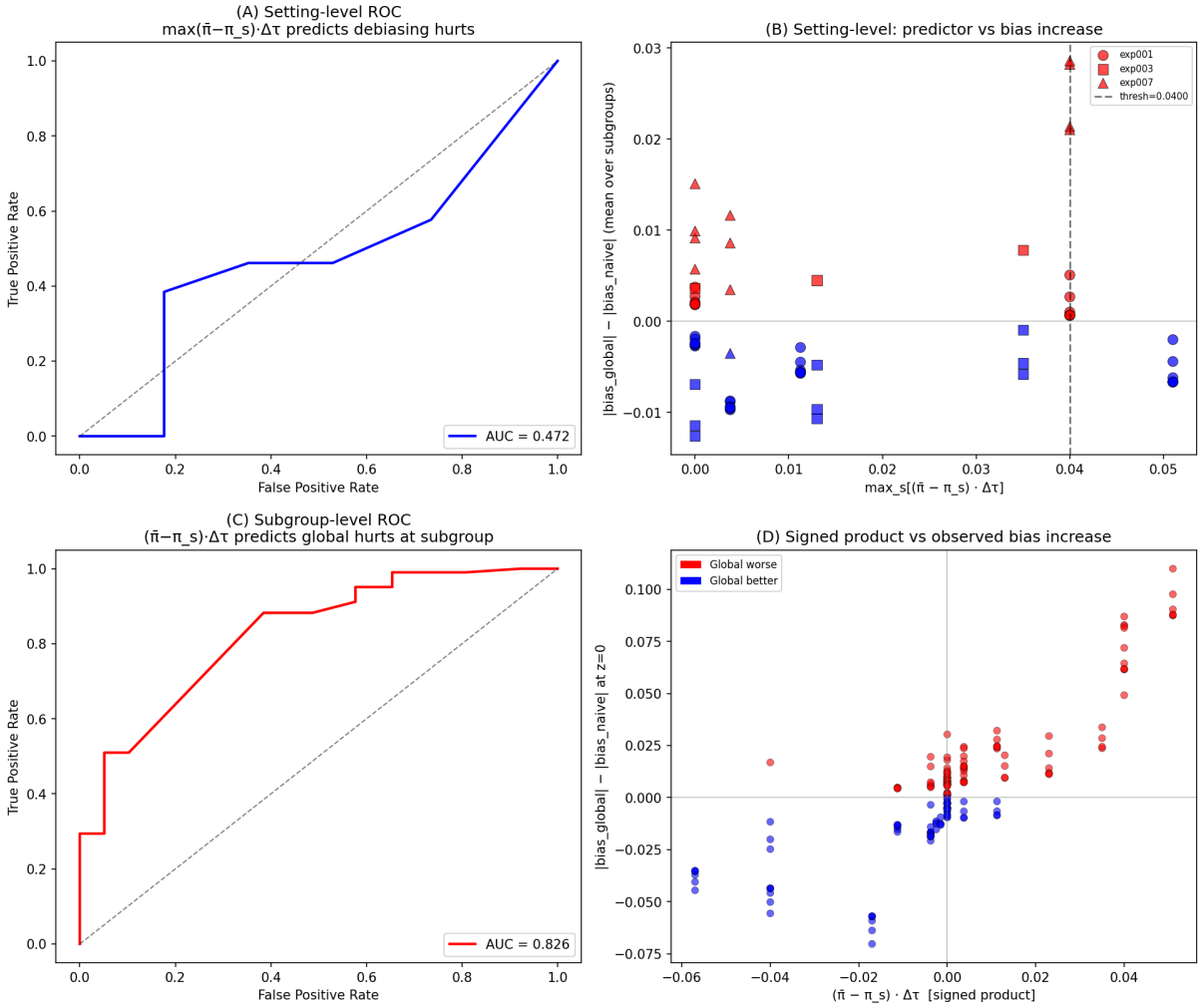


Figure 3: Diagnostic criterion performance. (A) Setting-level ROC (AUC $= 0.47$): aggregating across subgroups destroys the signal. (C) Subgroup-level ROC (AUC $= 0.83$): the $|\Delta\pi \times \Delta\tau|$ product reliably predicts which subgroups are harmed by global correction. (B, D) Scatter plots of the predictor against observed bias increase, colored by whether global correction helps or hurts.

bined with CausalForestDML (Section 4), with a computable diagnostic (AUC $= 0.833$) that identifies when stratified correction is warranted (Section 5.3). Composition with PPCI handles dual covariate/outcome error, reducing $|\text{bias}|$ from 0.179 to 0.007.

The recommended workflow is straightforward: compute $\Delta\pi \times \Delta\tau$ from a modest pilot sample (we use $n_{\text{expert}} = 50$ in our experiments); if the product is large for any subgroup, apply EC-HTE with $n_{\text{expert}} \geq 250$ (K=2) or $\geq 1{,}000$ (K=4); if uniformly small, global correction suffices. EC-HTE's advantage is conditional, not universal. When error heterogeneity is low, the variance cost of per-subgroup estimation outweighs the bias reduction, and global correction remains the appropriate choice. This conditionality is a feature of the framework: the diagnostic tells practitioners when to incur the cost of stratified correction and when not to.

Our validation uses symmetric misclassification and known subgroup structure. Extending the bias decomposition identity to asymmetric confusion matrices, continuous or latent subgroup discovery, and online settings with streaming annotations are natural next steps. As LLM annotation pipelines scale across the social sciences, the interaction between measurement error heterogeneity and causal effect heterogeneity will grow more consequential. The present work establishes the theoretical and empirical foundations for this correction in the discrete-subgroup, symmetric-error setting; the broader landscape of error-heterogeneity interactions in automated causal inference remains open.

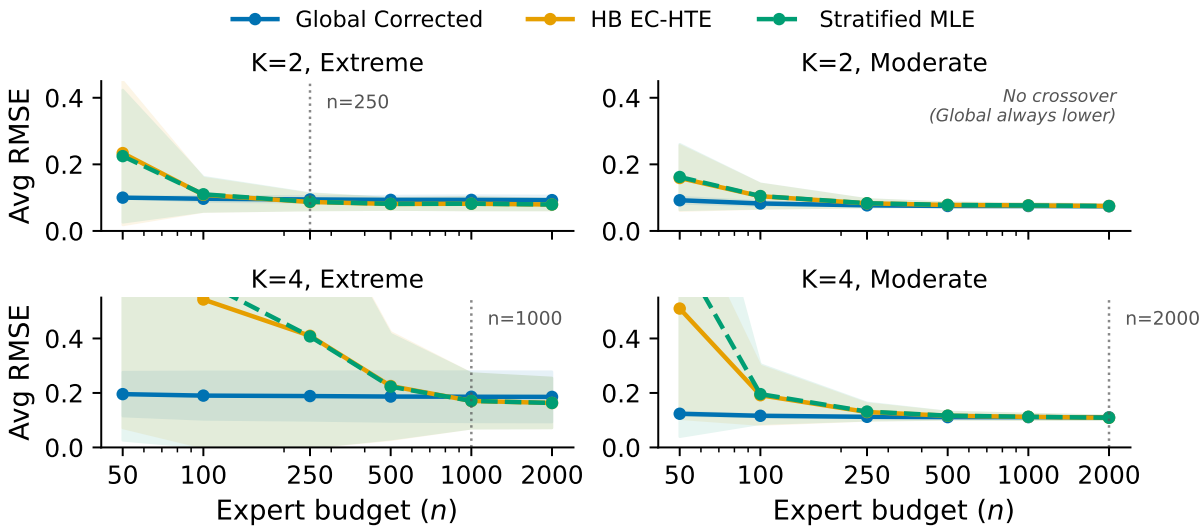


Figure 4: RMSE vs. expert budget (n_{expert}) across four regime$\times K$ combinations. Dashed vertical lines mark crossover points where EC-HTE achieves lower RMSE than global correction: n=250 (K=2, extreme), n=1000 (K=4, extreme). Under moderate heterogeneity (K=2), global correction dominates at all budgets. Shaded bands show ± 1 SE across 100 MC runs.

Limitations

Expert budget threshold. EC-HTE requires at least $n_{\text{expert}}/K \geq 25$ per subgroup for the hierarchical Bayesian shrinkage estimates to stabilize. Below this threshold, per-subgroup confusion matrix estimates are too noisy for reliable inversion, and stratified MLE is preferred. At $n_{\text{expert}} \geq 1000$, HB and stratified MLE converge to within 0.5% on all metrics, making the hierarchical model unnecessary at large expert budgets.

RMSE tradeoff at small n. EC-HTE achieves lower bias than global correction but incurs higher variance when the expert budget is small relative to the number of subgroups. In the most unfavorable configuration (K=4, extreme, $n_{\text{expert}} \leq 100$), EC-HTE CausalForest RMSE reaches 3.5$\times$ that of the naive estimator. Practitioners should weigh whether bias or variance is more consequential for their application before adopting subgroup-stratified correction at small sample sizes.

Plug-in standard errors. The variance-adjusted CausalForest propagates outcome uncertainty through the inverse confusion matrix $\hat{C}_s^{-1}$ but does not propagate uncertainty from the confusion matrix estimation itself. This plug-in approximation is reasonable at $n_{\text{expert}} \geq 100$, where confusion matrix estimation variance is small relative to outcome variance, but may understate total uncertainty below this threshold. Full Bayesian propagation of confusion matrix posterior uncertainty through the causal forest splitting procedure is computationally prohibitive and left to future work.

Symmetric misclassification assumption. All experiments use symmetric error rates ($P(\hat{Z}{=}1 \mid Z{=}0) = P(\hat{Z}{=}0 \mid Z{=}1)$). Real LLM annotators may exhibit asymmetric confusion patterns, such as higher false positive than false negative rates for certain label categories. The bias decomposition identity (Theorem 1) generalizes to asymmetric matrices, but empirical validation under asymmetric misclassification is left to future work.

Subgroup structure assumed known. EC-HTE requires discrete, pre-defined subgroups as input. Continuous effect modifiers or latent subgroup discovery settings require fundamentally different approaches. The hierarchical prior provides partial robustness to noisy subgroup definitions through automatic shrinkage toward the global mean, but intersectional subgroups with many levels may exceed the $n_{\text{expert}}/K \geq 25$ budget threshold, limiting practical applicability to settings with moderate K.

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Table 2: PPCI complementarity experiment (K=2, extreme, n_{expert}=250). DGP A: covariate error only; B: outcome error only (σ=0.5); C: both errors; D: no error. Values are |bias| and RMSE averaged over two subgroups. ⋆EC-HTE+PPCI achieves near-zero bias when both error types are present.

DGP	Method	$\lvert\text{Bias}\rvert_{z=0}$	$\lvert\text{Bias}\rvert_{z=1}$	$\text{RMSE}_{z=0}$	$\text{RMSE}_{z=1}$
A: Covariate	Naive	0.053	0.063	0.087	0.093
	Global	0.063	0.055	0.101	0.102
	EC-HTE	0.017	**0.013**	0.097	0.108
	PPCI	**0.015**	0.020	0.162	0.168
	EC-HTE+PPCI	0.017	**0.013**	0.097	0.108
B: Outcome	Naive	0.189	0.197	0.206	0.212
	Global	0.185	0.208	0.204	0.228
	EC-HTE	0.181	0.215	0.202	0.237
	PPCI	0.015	**0.020**	0.171	0.176
	EC-HTE+PPCI	**0.012**	0.023	0.171	0.186
C: Both	Naive	0.251	0.135	0.267	0.155
	Global	0.182	0.209	0.233	0.235
	EC-HTE	0.179	0.216	0.224	0.262
	PPCI	0.017	0.007	0.241	0.232
	EC-HTE+PPCI⋆	**0.007**	**0.008**	0.209	0.223
D: None	Naive	**0.007**	0.003	0.060	0.065
	Global	0.010	**0.002**	0.061	0.071
	EC-HTE	0.013	0.007	0.062	0.079
	PPCI	**0.007**	**0.003**	0.060	0.065
	EC-HTE+PPCI	0.013	0.007	0.062	0.079

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A Full Proof of Theorem 1

Setup. Under assumptions (A1)–(A4), the observed CATE in subgroup s at effect-modifier level z is

$$\tau_{\text{obs}}(z,s) = \sum_{z'} M_s[z,z'] \cdot \tau(z'), \tag{9}$$

where $M_s[z,z'] = P(Z{=}z' \mid \hat{Z}{=}z, S{=}s)$ is the mixing matrix. Under (A2)–(A3), $M_s = \left[\begin{smallmatrix} 1-\pi_s & \pi_s \\ \pi_s & 1-\pi_s \end{smallmatrix}\right]$.

Global correction. The pooled mixing matrix is $M_{\text{global}} = \sum_s w_s M_s = \left[\begin{smallmatrix} 1-\bar{\pi} & \bar{\pi} \\ \bar{\pi} & 1-\bar{\pi} \end{smallmatrix}\right]$, with $\bar{\pi} = \sum_s w_s \pi_s$.

Global correction applies M_{global}^{-1} to $\tau_{\text{obs}}(\cdot, s)$:

$$\hat{\tau}_{\text{global}}(z,s) = [M_{\text{global}}^{-1} M_s \cdot \boldsymbol{\tau}]_z, \tag{10}$$

where $\boldsymbol{\tau} = [\tau(0), \tau(1)]^\top$.

Bias computation. The bias is $\hat{\tau}_{\text{global}}(z,s) - \tau(z) = [(M_{\text{global}}^{-1} M_s - I)\boldsymbol{\tau}]_z$.

Direct computation gives:

$$M_{\text{global}}^{-1} = \frac{1}{1-2\bar{\pi}} \begin{bmatrix} 1-\bar{\pi} & -\bar{\pi} \\ -\bar{\pi} & 1-\bar{\pi} \end{bmatrix}, \tag{11}$$

$$M_{\text{global}}^{-1} M_s = \frac{1}{1-2\bar{\pi}} \begin{bmatrix} (1-\bar{\pi})(1-\pi_s) - \bar{\pi}\pi_s & (1-\bar{\pi})\pi_s - \bar{\pi} \\ -\bar{\pi}(1-\pi_s) + (1-\bar{\pi})\pi_s & -\bar{\pi}\pi_s + (1-\bar{\pi} \end{bmatrix} \tag{12}$$

Simplifying the $(1,1)$ entry: $(1-\bar{\pi})(1-\pi_s) - \bar{\pi}\pi_s = 1 - \bar{\pi} - \pi_s + \bar{\pi}\pi_s - \bar{\pi}\pi_s = 1 - \bar{\pi} - \pi_s + 2\bar{\pi}\pi_s - \bar{\pi}\pi_s$.

More directly, define $\delta_s = \pi_s - \bar{\pi}$. Then:

$$M_{\text{global}}^{-1} M_s - I = \frac{\delta_s}{1-2\bar{\pi}} \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}. \tag{13}$$

Applying to $\boldsymbol{\tau}$:

$$\text{bias}(z{=}0, s) = \frac{\delta_s}{1-2\bar{\pi}}(-\tau(0) + \tau(1)) = +\frac{\delta_s \cdot \Delta\tau}{1-2\bar{\pi}}, \tag{14}$$

$$\text{bias}(z{=}1, s) = \frac{\delta_s}{1-2\bar{\pi}}(\tau(0) - \tau(1)) = -\frac{\delta_s \cdot \Delta\tau}{1-2\bar{\pi}}, \tag{15}$$

where $\Delta\tau = \tau(1) - \tau(0)$. □

Proof of Proposition 3. The naive estimator has bias $\tau_{\text{obs}}(z,s) - \tau(z)$. Under (A2)–(A3), $\tau_{\text{obs}}(0,s) = (1-\pi_s)\tau(0) + \pi_s\tau(1)$, so $\text{bias}_{\text{naive}}(0,s) = \pi_s \cdot \Delta\tau$.

Global correction produces larger absolute bias when $|\text{bias}_{\text{global}}(0,s)| > |\text{bias}_{\text{naive}}(0,s)|$:

$$\frac{|\delta_s| \cdot |\Delta\tau|}{|1-2\bar{\pi}|} > \pi_s \cdot |\Delta\tau|. \tag{16}$$

Since $\Delta\tau \neq 0$ (otherwise both biases are zero), dividing through gives $|\delta_s|/((1-2\bar{\pi}) \cdot \pi_s) > 1$, where we use $1 - 2\bar{\pi} > 0$ from (A4). □